

Shallow water analysis

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Corrections in red.

Green: the cross-channel superelevation pressure $-g \partial_n h'$ (non-dimensional coefficient $\varepsilon_2/(\varepsilon_1 Fr^2 \alpha^2) \sim 1/(Fr^2 \alpha^2) \geq 1$) — the shallow-water term that can never be small.

1 Shallow water equation — the fast (flow) timescale

We write the shallow water equation

$$\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} = -g \nabla \eta - \frac{C_f |\mathbf{u}| \mathbf{u}}{h}, \quad (1)$$

$$\partial_t h + \nabla \cdot (\mathbf{u} h) = 0. \quad (2)$$

Let z_b be the bed elevation, so that $h = \eta - z_b$ and $\nabla \eta = \nabla h + \nabla z_b$. We write it in channel-following coordinate:

$$\partial_t u + u \partial_s u + v \partial_n u + C u v = -g \partial_s (h + z_b) - \frac{C_f |\mathbf{u}|}{h} u, \quad (3)$$

$$\partial_t v + u \partial_s v + v \partial_n v - C u^2 = -g \partial_n (h + z_b) - \frac{C_f |\mathbf{u}|}{h} v, \quad (4)$$

$$\partial_t h + \partial_s (u h) + \partial_n (v h) + C v h = 0. \quad (5)$$

Assume $u = U_0 + u'$, $v = v'$, $C = C'$, $h = H_0 + h'$, and assume that $u'/U_0 \sim h'/H_0 \sim \mathcal{O}(\varepsilon)$ ($z_b \sim H_0 = \mathcal{O}(1)$: the bed sits in the base state ($H_0 + z_b$), not the perturbation — only its secondary-flow tilt z'_b of §3 is $\mathcal{O}(\varepsilon)$). If we neglect all $\mathcal{O}(\varepsilon^2)$ terms, we have

$$\begin{aligned} \partial_t u' + U_0 \partial_s U_0 + U_0 \partial_s u' + u' \partial_s U_0 + v' \partial_n U_0 \\ = -g \partial_s (H_0 + z_b) - g \partial_s h' - \frac{C_f U_0^2}{H_0} - \frac{2C_f U_0 u'}{H_0} + \frac{C_f U_0^2}{H_0^2} h' \end{aligned} \quad (6)$$

$$\partial_t v' + U_0 \partial_s v' - C' U_0^2 = -g \partial_n (H_0 + z_b) - g \partial_n h' - \frac{C_f U_0 v'}{H_0} \quad (7)$$

$$\partial_t h' + \partial_s (U_0 H_0) + \partial_s (U_0 h') + \partial_s (u' H_0) + \partial_n (v' H_0) = 0 \quad (8)$$

Leading order balance:

$$U_0 \partial_s U_0 = -g \partial_s (H_0 + z_b) - \frac{C_f U_0^2}{H_0}, \quad (9)$$

$$-g \partial_n (H_0 + z_b) = 0, \quad (10)$$

$$\partial_s (U_0 H_0) = 0. \quad (11)$$

With $I \equiv -\partial_s z_b$ the valley slope, (9)–(11) are satisfied by $H_0(n) + z_b(n) = \text{const}$ together with $C_f U_0(n)^2 / H_0(n) = gI$, that is

$$U_0(n) = U_0 \tilde{\xi}(\tilde{n}), \quad H_0(n) = H_0 \tilde{\xi}(\tilde{n})^2, \quad C_f U_0^2 / H_0 = gI.$$

The depth profile is therefore slaved to the velocity profile: $H_0 \propto \tilde{\xi}^2$. A quadratic jet $\tilde{\xi} = 1 + r(1 - \tilde{n}^2)$ hence forces a *quartic* bed. The two profiles cannot be prescribed independently.

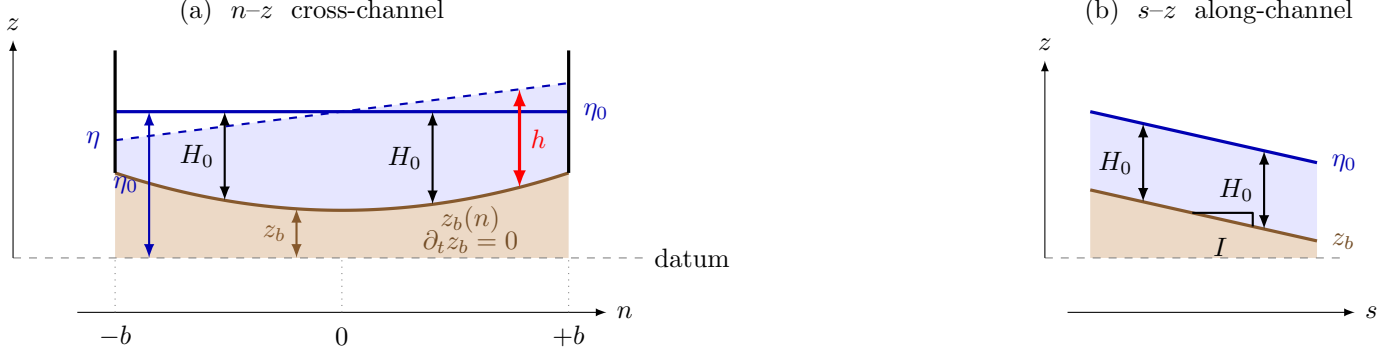


Figure 1: Elevations are measured from the datum; depths are the gaps between them. η_0 = base free-surface elevation, z_b = bed elevation, $H_0 = \eta_0 - z_b$ = base depth, $h = \eta - z_b = H_0 + h'$ = actual depth. (a) The bed is non-flat in n ; (10) makes η_0 flat across the section, so $H_0(n) = H_0 \tilde{\xi}(\tilde{n})^2$ varies with n through the bed alone. The dashed surface $\eta = \eta_0 + h'$ is the perturbation; the gap between the two blue lines is h' . (b) z_b and η_0 drop together at the valley slope $I = -\partial_s z_b$, and $-g\partial_s(h + z_b) = -g\partial_s \eta = gI$ drives the flow. Note $\partial_s z_b \neq 0$ — the bed elevation must fall downstream — while H_0 and the cross-sectional shape are the same at every s . “The bed does not change” means $\partial_t z_b = 0$, not $\partial_s z_b = 0$.

On the fast time $T_f = \Lambda / U_0$ the flow equilibrates to the (fixed) banks, so we treat it as *steady*, $\partial_t \rightarrow 0$; the slow bank motion is restored in §2. We non-dimensionalize the $\mathcal{O}(\varepsilon)$ equation with

$$\tilde{u} = u' / U, \quad \tilde{v} = v' / V, \quad \tilde{s} = s / \Lambda, \quad \tilde{n} = n / b, \quad C' = C \tilde{C}(s), \quad \tilde{h} = h' / \mathcal{H}, \quad \Lambda = \lambda / 2\pi, \quad (12)$$

we have

$$\frac{U U_0}{\Lambda} \tilde{\xi} \tilde{\partial}_s \tilde{u} + \frac{V U_0}{b} \tilde{v} \tilde{\partial}_n \tilde{\xi} = -\frac{g \mathcal{H}}{\Lambda} \tilde{\partial}_s \tilde{h} - \frac{2 C_f U_0 U}{H_0} \frac{\tilde{u}}{\tilde{\xi}} + \frac{C_f U_0^2 \mathcal{H}}{H_0^2} \frac{\tilde{h}}{\tilde{\xi}^2}, \quad (13)$$

$$\frac{V U_0}{\Lambda} \tilde{\xi} \tilde{\partial}_s \tilde{v} - C U_0^2 \tilde{C} \tilde{\xi}^2 = -\frac{g \mathcal{H}}{b} \tilde{\partial}_n \tilde{h} - \frac{C_f U_0 V}{H_0} \frac{\tilde{v}}{\tilde{\xi}}, \quad (14)$$

$$\frac{\mathcal{H} U_0}{\Lambda} \tilde{\xi} \tilde{\partial}_s \tilde{h} + \frac{H_0 U}{\Lambda} \tilde{\partial}_s (\tilde{\xi}^2 \tilde{u}) + \frac{H_0 V}{b} \tilde{\partial}_n (\tilde{\xi}^2 \tilde{v}) = 0. \quad (15)$$

Define

$$\alpha = b / \Lambda, \quad \gamma = V / U, \quad \varepsilon_1 = U / U_0, \quad \varepsilon_2 = \mathcal{H} / H_0, \quad Fr = U_0 / \sqrt{g H_0}, \quad F_c = C_f \Lambda / H_0, \quad Ci = C b \quad (16)$$

we have

$$\tilde{\xi} \tilde{\partial}_s \tilde{u} + \frac{\gamma}{\alpha} \tilde{v} \tilde{\partial}_n \tilde{\xi} = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_s \tilde{h} - 2 F_c \frac{\tilde{u}}{\tilde{\xi}} + F_c \frac{\varepsilon_2}{\varepsilon_1} \frac{\tilde{h}}{\tilde{\xi}^2}, \quad (17)$$

$$\tilde{\xi} \tilde{\partial}_s \tilde{v} - \frac{Ci}{\varepsilon_1 \alpha \gamma} \tilde{C} \tilde{\xi}^2 = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2 \alpha \gamma} \tilde{\partial}_n \tilde{h} - F_c \frac{\tilde{v}}{\tilde{\xi}}, \quad (18)$$

$$\tilde{\partial}_s(\tilde{\xi}^2 \tilde{u}) + \frac{\gamma}{\alpha} \tilde{\partial}_n(\tilde{\xi}^2 \tilde{v}) + \frac{\varepsilon_2}{\varepsilon_1} \tilde{\xi} \tilde{\partial}_s \tilde{h} = 0. \quad (19)$$

The green term is the cross-channel superelevation pressure: the free surface tilts across the channel to balance the centrifugal forcing $\propto Ci \tilde{C}$, and its streamwise gradient sets the near-bank u'_b (Ikeda's $-U_0^2 \partial_s \mathcal{C}'$). Its coefficient $\varepsilon_2/(\varepsilon_1 Fr^2 \alpha \gamma) \sim 1/(Fr^2 \alpha^2)$ is ≥ 1 for every subcritical ($Fr < 1$), long-wave ($\alpha < 1$) meander — $O(1/\varepsilon)$ in limit 1, $O(1)$ in limit 2 — so it can never be small. That is why limit 2 (the QGPV limit, which *keeps* it) still migrates downstream. Discarding the free surface entirely (rigid lid, no $\tilde{h}$) is the only way to remove it; the pure-vorticity model that then remains migrates upstream on its own β -Rossby dynamics.

We typically assume $\alpha = \gamma$. If we assume $Ci \sim \varepsilon_1 \sim \varepsilon_2 \sim \mathcal{O}(\varepsilon)$, $\alpha \sim \mathcal{O}(\varepsilon^{1/2})$, $Fr \sim F_c \sim 1$, the leading order equation becomes

$$\tilde{\xi} \tilde{\partial}_s \tilde{u} + \tilde{v} \tilde{\partial}_n \tilde{\xi} = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_s \tilde{h} - 2F_c \frac{\tilde{u}}{\tilde{\xi}} + F_c \frac{\varepsilon_2}{\varepsilon_1} \frac{\tilde{h}}{\tilde{\xi}^2}, \quad (20)$$

$$\frac{1}{\varepsilon_1 \alpha^2} Ci \tilde{C} \tilde{\xi}^2 = \frac{1}{\varepsilon_1 \alpha^2} \frac{\varepsilon_2}{Fr^2} \tilde{\partial}_n \tilde{h}, \quad (21)$$

$$\tilde{\partial}_s(\tilde{\xi}^2 \tilde{u}) + \tilde{\partial}_n(\tilde{\xi}^2 \tilde{v}) + \frac{\varepsilon_2}{\varepsilon_1} \tilde{\xi} \tilde{\partial}_s \tilde{h} = 0. \quad (22)$$

Equations 20 and Equation 21 lead to the solution in Ikeda et al.

If we assume that $\varepsilon_2 \sim \mathcal{O}(\varepsilon^2)$, $Ci \sim \varepsilon_1 \sim Fr^2 \sim \mathcal{O}(\varepsilon)$, and set $\alpha = 1$ (so $Ci/\varepsilon_1 = O(1)$: the curvature forcing survives — a *forced* QGPV, as in §1.1), the leading order equation becomes

$$\tilde{\xi} \tilde{\partial}_s \tilde{u} + \tilde{v} \tilde{\partial}_n \tilde{\xi} = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_s \tilde{h} - 2F_c \frac{\tilde{u}}{\tilde{\xi}}, \quad (23)$$

$$\tilde{\xi} \tilde{\partial}_s \tilde{v} - \frac{Ci}{\varepsilon_1 \alpha^2} \tilde{C} \tilde{\xi}^2 = -\frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_n \tilde{h} - F_c \frac{\tilde{v}}{\tilde{\xi}}, \quad (24)$$

$$\tilde{\partial}_s(\tilde{\xi}^2 \tilde{u}) + \tilde{\partial}_n(\tilde{\xi}^2 \tilde{v}) = 0. \quad (25)$$

Equations 23-25 leads to the QGPV equation

$$\tilde{\xi} \tilde{\partial}_s \tilde{\zeta} + \tilde{\beta} \tilde{v} = -\frac{F_c}{\tilde{\xi}} \tilde{\zeta} + \frac{F_c}{\tilde{\xi}} \tilde{\partial}_n \tilde{u} - \frac{2F_c \tilde{u}}{\tilde{\xi}^2} \tilde{\partial}_n \tilde{\xi} + \frac{Ci}{\varepsilon_1 \alpha^2} \tilde{\xi}^2 \tilde{\partial}_s \tilde{C}, \quad (26)$$

where

$$\tilde{\beta} = 2 \frac{(\tilde{\partial}_n \tilde{\xi})^2}{\tilde{\xi}} - \tilde{\partial}_{nn} \tilde{\xi}, \quad \tilde{\zeta} = \tilde{\partial}_s \tilde{v} - \tilde{\partial}_n \tilde{u}, \quad \tilde{\xi}^2 \tilde{u} = -\tilde{\partial}_n \tilde{\psi}, \quad \tilde{\xi}^2 \tilde{v} = \tilde{\partial}_s \tilde{\psi}. \quad (27)$$

With the *flat-bed* choice $H_0 = \text{const}$ (rather than the balanced $H_0 = H_0 \tilde{\xi}^2$ derived at (9)–(11)), the friction denominators lose a power of $\tilde{\xi}$ ($\tilde{u}/\tilde{\xi} \rightarrow \tilde{\xi} \tilde{u}$), continuity (25) becomes the non-divergent $\tilde{\partial}_s \tilde{u} + \tilde{\partial}_n \tilde{v} = 0$, and $\tilde{\beta} \rightarrow -\tilde{\partial}_{nn} \tilde{\xi}$; equations (23)–(27) then coincide term-for-term with the July-24 update. The slaved-bed form above ($H_0 = \tilde{\xi}^2$) is the self-consistent one — its base state actually exists — and is what the Thetis model solves.

We can put:

$$\tilde{\xi}(\tilde{n}) \tilde{\partial}_s \tilde{u} + \tilde{v} \tilde{\partial}_n \tilde{\xi}(\tilde{n}) + \frac{\varepsilon_2}{\varepsilon_1 Fr^2} \tilde{\partial}_s \tilde{h} + 2F_c \frac{\tilde{u}}{\tilde{\xi}(\tilde{n})} - F_c \frac{\varepsilon_2}{\varepsilon_1} \frac{\tilde{h}}{\tilde{\xi}(\tilde{n})^2} = 0, \quad (28)$$

$$\tilde{\xi}(\tilde{n}) \tilde{\partial}_s \tilde{v} - \frac{Ci}{\varepsilon_1 \alpha^2} \tilde{C}(\tilde{s}) \tilde{\xi}(\tilde{n})^2 + \frac{\varepsilon_2}{\varepsilon_1 Fr^2 \alpha^2} \tilde{\partial}_n \tilde{h} + F_c \frac{\tilde{v}}{\tilde{\xi}(\tilde{n})} = 0, \quad (29)$$

$$\tilde{\partial}_s(\tilde{\xi}(\tilde{n})^2 \tilde{u}) + \tilde{\partial}_n(\tilde{\xi}(\tilde{n})^2 \tilde{v}) + \frac{\varepsilon_2}{\varepsilon_1} \tilde{\xi}(\tilde{n}) \tilde{\partial}_s \tilde{h} = 0. \quad (30)$$

From gravity to Rossby: the curl and its ordering

Equations (17)–(19) carry seven groups $\alpha, \gamma, \varepsilon_1, \varepsilon_2, Fr, F_c, Ci$ (with $\alpha = \gamma$); their relative orders decide whether the leading balance is the gravity/Ikeda response or a Rossby wave. The Rossby branch lives on the *vorticity* equation, obtained by the curl $\tilde{\partial}_s(18) - \tilde{\partial}_n(17)$. Write $P_s = \varepsilon_2/(\varepsilon_1 Fr^2)$ (the $\tilde{\partial}_s \tilde{h}$ coefficient in (17)) and $P_n = \varepsilon_2/(\varepsilon_1 Fr^2 \alpha \gamma) = P_s/(\alpha \gamma)$ (the $\tilde{\partial}_n \tilde{h}$ coefficient in (18)); the two pressure terms combine as

$$\tilde{\partial}_s(P_n \tilde{\partial}_n \tilde{h}) - \tilde{\partial}_n(P_s \tilde{\partial}_s \tilde{h}) = (P_n - P_s) \tilde{\partial}_s \tilde{\partial}_n \tilde{h} = P_s \left(\frac{1}{\alpha \gamma} - 1 \right) \tilde{\partial}_s \tilde{\partial}_n \tilde{h},$$

which vanishes *identically* iff $P_n = P_s$, i.e. $\boxed{\alpha \gamma = 1}$, hence $\alpha = \gamma = 1$: the pressure cancels in the curl. This is geometric and Fr -independent, and it is a *coefficient match*, not a smallness — with $Fr^2 \sim \varepsilon_1$ the pressure $P_s = O(1)$ is a leading-order term of the un-curled momentum equations; it leaves the *vorticity* budget only because $\alpha = 1$ equalises the two length scales Λ, b . When $\alpha \ll 1$ (Ikeda) $P_n = P_s/\alpha^2 \gg P_s$: the $\tilde{\partial}_n \tilde{h}$ pressure dominates (18) — cross-channel superelevation, a gravity response with no vorticity dynamics.

With $\alpha = \gamma = 1$, continuity (19) is non-divergent provided $\varepsilon_2/\varepsilon_1 \ll 1$ (rigid lid), so a stream-function exists ($\tilde{\xi}^2 \tilde{u} = -\tilde{\partial}_n \tilde{\psi}$, $\tilde{\xi}^2 \tilde{v} = \tilde{\partial}_s \tilde{\psi}$) and $\tilde{\partial}_s \tilde{u} + \tilde{\partial}_n \tilde{v} = -2(\tilde{\partial}_n \tilde{\xi}/\tilde{\xi}) \tilde{v}$. Substituting into the curled inertia gives the jet's background PV gradient

$$\tilde{\beta} = \frac{2(\tilde{\partial}_n \tilde{\xi})^2}{\tilde{\xi}} - \tilde{\partial}_{nn} \tilde{\xi} = \tilde{\xi}^2 \tilde{\partial}_n q_{\text{base}}, \quad q_{\text{base}} = -\frac{\tilde{\partial}_n \tilde{\xi}}{\tilde{\xi}^2}$$

(base velocity $\tilde{\xi}$, base depth $\tilde{\xi}^2$). The $+2(\tilde{\partial}_n \tilde{\xi})^2/\tilde{\xi}$ term is the depth-slaving signature — a constant-depth base drops it and leaves the wrong $-\tilde{\partial}_{nn} \tilde{\xi}$; a uniform jet has $\tilde{\beta} = 0$ and no wave. The leading-order vorticity equation is

$$\tilde{\xi} \tilde{\partial}_s \tilde{\zeta} + \tilde{\beta} \tilde{v} + \frac{F_c}{\tilde{\xi}} [\tilde{\zeta} - \tilde{\xi}^2 \tilde{\partial}_n (\tilde{u}/\tilde{\xi}^2)] = \frac{Ci}{\varepsilon_1} \tilde{\xi}^2 \tilde{\partial}_s \tilde{C}, \quad \tilde{\zeta} = \tilde{\partial}_s \tilde{v} - \tilde{\partial}_n \tilde{u} \quad (\star)$$

— advection of vorticity along the jet, the shear-Rossby restoring $\tilde{\beta} \tilde{v}$, quadratic bottom-drag (the factor-2 asymmetry $2F_c \tilde{u}/\tilde{\xi}$ vs $F_c \tilde{v}/\tilde{\xi}$), and the curvature forcing; exactly the $U q'_x + v' Q_y = \text{friction}$ form.

The orderings that select each branch:

parameter	Rossby / QGPV	Gravity / Ikeda	role
$\alpha = \gamma$	$O(1)$ [=1]	$O(\varepsilon^{1/2})$	pressure cancels vs dominates
$\varepsilon_2/\varepsilon_1$	$O(\varepsilon)$	$O(1)$	(non-)divergent
Fr^2	$O(\varepsilon)$	$O(1)$	low- vs finite-Froude
Ci/ε_1	$O(1)$ forced	$O(1)$	curvature source
F_c	$O(1)$	$O(1)$	friction leading order

The single switch is $\alpha \gamma = 1$ Rossby (pressure curls away), $\ll 1$ gravity (pressure dominates (18)); low Fr is the consistent secondary marker. Forced vs free is Ci/ε_1 ($O(1)$ keeps the source through $\tilde{\partial}_s \tilde{C}$; $\ll 1$ leaves the free shear-Rossby wave).

2 Slow timescale: bank migration

Section 1's flow is steady on the fast time $T_f = \Lambda/U_0$: given the banks it equilibrates within a few advection times. The banks themselves move on a much slower time set by the erosion law.

Let $y_b(s, t)$ be a bank's lateral position and $u'_b = u'(s, \pm b)$ the near-bank velocity from §1; with $\psi'_1, \psi'_2, \psi'_3$ the streamfunction perturbation at $n = +b, 0, -b$,

$$\psi'_1 = \bar{u}(b) y_b = U_0 y_b, \quad (31)$$

$$u'_b = (\psi'_2 - \psi'_1)/b, \quad (32)$$

$$\partial_t \psi'_1 = \frac{\varepsilon C_f U_0}{b} (\psi'_2 - \psi'_1), \quad (33)$$

$$\gamma \partial_t y_b = E u'_b, \quad E = \varepsilon C_f, \quad \gamma = \cos \theta. \quad (34)$$

Equation (33) is the three-level form; (34) is Ikeda, Parker & Sawai (1981) eqs (11)–(13). For two independently mobile banks y_N, y_S ,

$$\partial_t y_N = +E_{e,d} u'_N, \quad \partial_t y_S = -E_{e,d} u'_S, \quad (35)$$

with E_e where $u' > 0$ (erosion) and E_d where $u' < 0$ (deposition); $E_e = E_d$ gives constant width.

The slow timescale. How long does a bank take to move? Scale $y_b \sim b$ and $u'_b \sim U = \varepsilon_1 U_0$, and rescale time $t = T_s \tau$ in (34):

$$\gamma \frac{b}{T_s} \partial_\tau \tilde{y}_b = \varepsilon C_f U \tilde{u}'_b \implies T_s = \frac{\gamma b}{\varepsilon C_f U} = \frac{\gamma b}{\varepsilon C_f \varepsilon_1 U_0}, \quad (36)$$

chosen so the bank evolves at $O(1)$, $\partial_\tau \tilde{y}_b = \tilde{u}'_b$. The flow and the bank are thus non-dimensionalized by *different* time parameters — $T_f = \Lambda/U_0$ (advection) and T_s (erosion) — with ratio

$$\frac{T_f}{T_s} = \frac{\varepsilon C_f \varepsilon_1}{\gamma \alpha} \ll 1, \quad \alpha = \frac{b}{\Lambda}. \quad (37)$$

This ratio — the product of the erosion coefficient εC_f and the flow perturbation ε_1 , *not* the ε of §1 — is the genuine small parameter of the morphodynamics. Because $T_f/T_s \ll 1$, the ∂_t acting on the flow is negligible: this is precisely why §1 is steady, the flow adiabatically following the instantaneous planform while all time evolution lives on the bank. The meander grows and migrates on T_s .

3 Secondary flow and the bed — the A switch

The note has no secondary-flow term; a depth-averaged 2-D model cannot generate the helical circulation of a bend, so its effect on the bed is *imposed* as a curvature-slaved perturbation to the frozen depth of §1. Writing the total depth as $H(s, n) = H_0(n) + H'_2(s, n)$ with $H_0(n)$ the time-independent base profile, Ikeda, Parker & Sawai (1981) eq. (6) is

$$H'_2(s, n) = -A H_0(n) \mathcal{C}'(s) n, \quad n = b\tilde{n}, \quad (38)$$

a pool ($H'_2 > 0$, deeper) on the outer bank and a point bar on the inner bank, both proportional to the local centreline curvature $\mathcal{C}'(s)$. (This is Ikeda's bed perturbation η' , renamed to avoid clashing with the free surface η ; the sign is fixed so the pool is on the outer bank. It enters the bed of §1 as $z_b = \bar{z}_b(n) + z'_b$ with $z'_b = -H'_2$.) The secondary-flow strength is a switch,

$$A = 0 : \text{incised, no secondary flow } (H'_2 \equiv 0, \text{ frozen bed}); \quad A = 2.89 : \text{alluvial (Suga 1963)}. \quad (39)$$

$H_0(n)$ never changes in time; H'_2 carries no independent dynamics — it is slaved to the slowly migrating planform (recomputed from the current curvature each morphological step), so “the bed does not change” still means $\partial_t z_b = 0$. With $h' = \xi' - \eta'$ and Ikeda’s superelevation $\xi' = C' U_0^2 n / g$, the depth perturbation is $h' / H_0 = (F^2 + A) C' \tilde{n}$: A and F^2 occupy the *same* depth/drag slot, so $A = 0$ leaves only $F^2 \approx 0.09$ while $A = 2.89$ raises the bend forcing to ≈ 2.98 — which is what turns bend decay into growth.

4 Boundary conditions

The note is an interior linear analysis and states no boundary conditions; the model closes the channel $[0, L] \times [-b, b]$ with

$$\underbrace{u_n = -\bar{u}(n)}_{\text{inflow } s=0}, \quad \underbrace{\eta = \eta_{\text{ref}} - I L}_{\text{outflow } s=L}, \quad \underbrace{u_n = 0}_{\text{banks } n=\pm b} . \quad (40)$$

The flow is subcritical ($Fr < 1$), so exactly one characteristic enters at each open boundary: the inflow prescribes the velocity profile only, the outflow the elevation only. Pinning η , u and v together at one open boundary would over-determine it (and, started from rest, launch a spurious bore). The banks are free-slip — no normal flux, no wall drag beyond the bed friction. Drag is quadratic, $-C_f |\mathbf{u}| \mathbf{u} / h$ with $C_f = 0.05$; a small horizontal viscosity $\nu = 0.05 \text{ m}^2 \text{ s}^{-1}$ is retained for DG stability (both stated here because the note carries neither).